OCC: SOFTWARE ENGINEER YEAR 2

SAQA ID
NQF 6
Software design and testing C#
SDT621
WEEK 1 INDUSTRIAL PROJECT
2026

CTU training solutions | Private Tertiary Education

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DECLARATION OF AUTHENTICITY

I Zander du Plessis (FULL NAME)

hereby declare that the contents of this assessment SDT 621

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: _____

Date: 24/04/26

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QUALIFICATION AND PROJECT DETAILS

Programme Title	OCC: SOFTWARE ENGINEER
Module Title	Software design and testing C#
Module Code	SDT621
NQF Level	6
Assessment Type	WEEK 1 INDUSTRIAL PROJECT
Hand Out Date	17 / 0 4 / 2026
Project Developer	Lusukama Selemani
Submission code via	Github

Student Name and Surname	Zander du Plessis
Student Number	20240028

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Instructions:

Please read the following instructions carefully before attempting this project assessment.

General Instructions

- Read the full project scenario and all tasks carefully before starting your work.
- Pay attention to the requirements for each section of the project.
- Ensure that you answer all required tasks in both the Console Application and the Windows Forms Application components.
- Apply the C# programming concepts covered in Week 1, including:
  - variables and data types
  - identifiers and keywords
  - operators and expressions
  - methods
  - scope (global and local variables)
  - type casting
  - user input and output
  - basic Windows Forms interface design
- Use meaningful variable names and proper coding conventions.
- Ensure your program produces clear and well-formatted output.
- Test your application before submission to confirm that it runs correctly without errors.

GitHub Submission Requirement

You are required to:

- Push your project code to GitHub
- Ensure the repository contains the correct and complete solution
- Share the GitHub repository link in your submission for review and verification
- Example: https://github.com/your-username/project-name

Make sure your repository:

- builds successfully
- contains all project files
- matches the work submitted in your ZIP file

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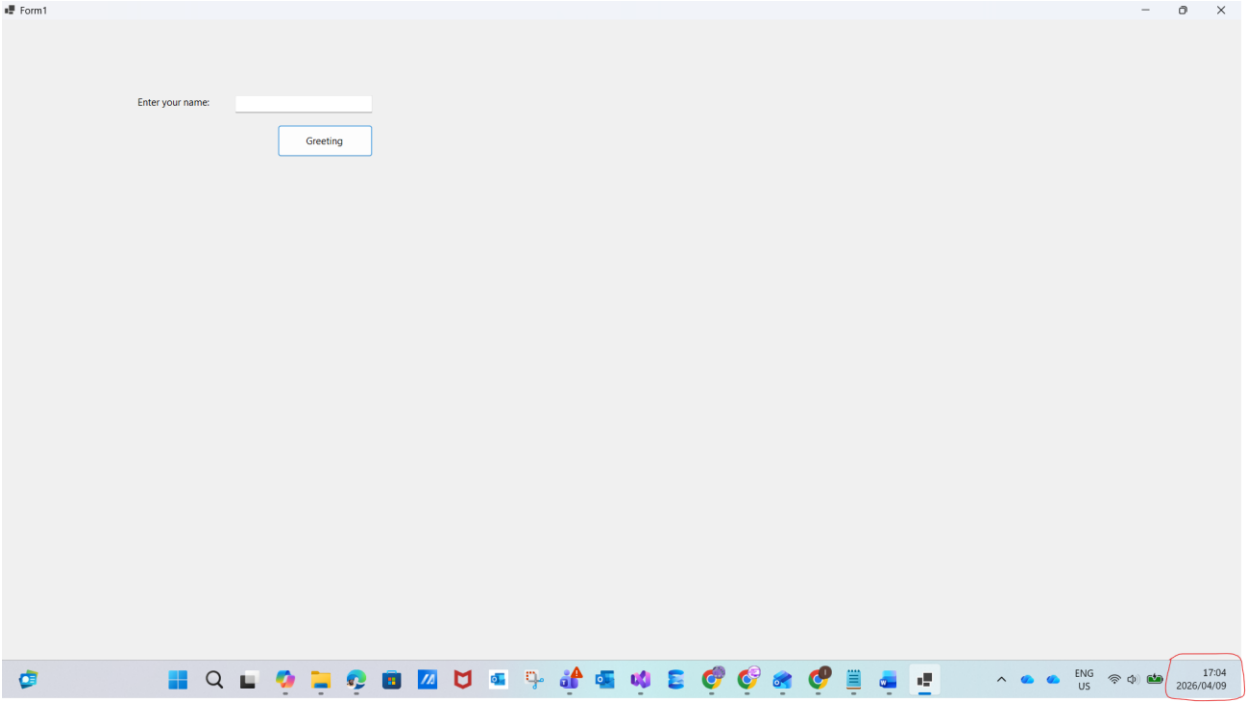
Learning Outcomes:

- Structure of C# program
- Variables & data types
- Identifiers & keywords
- Arithmetic operators
- Expressions
- Methods
- Method parameters
- Scope (global/local)
- Type casting
- Increment / decrement
- Console input/output
- Windows Forms interface basics

Example of Collecting Evidence for your submission pdf:

Question1 :



Ensure each screenshot shows the **full screen**, including the **system date and time**

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Smart Student Profile Processor (Console + Windows Forms Starter Version)

Project Overview

You are required to develop a C# project-based solution based on the Week 1 content covered in class. This task is designed to simulate a real workplace activity where a junior developer is asked to build a simple student information processing system.

The project must demonstrate your understanding of the following Week 1 concepts: creating a basic C# program, using Visual Studio, variables, data types, operators, methods, scope, and basic Windows Forms development. These topics are part of the Week 1 outcomes in the provided slide content.

Project Purpose

You are hired as a Junior Software Developer at Kitumaini Digital Solutions.

The company is building a Student Readiness Capture System used by training providers to:

- capture student information
- validate data
- calculate readiness indicators
- display structured outputs
- prepare data for future database storage

Your responsibility this week:

Develop Version 1 (Console Application + Basic Windows Form) of the system.

Key Focus Areas

Your submission will be evaluated based on:

- correct program structure
- correct use of variables and data types
- correct use of operators
- correct use of methods
- implementation of method overloading
- correct scope usage
- input validation
- Windows Forms interface functionality
- program output formatting

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